

RAG-Based Customer Support Assistant

Technical Documentation

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1. Introduction

This project implements a Retrieval-Augmented Generation (RAG) system that answers user queries using a PDF knowledge base. It combines retrieval and response generation and includes a Human-in-the-Loop (HITL) mechanism for handling uncertain queries.

2. System Overview

The system works as follows:

1. Load PDF document
 2. Split text into chunks
 3. Convert chunks into embeddings
 4. Store embeddings in ChromaDB
 5. Accept user query
 6. Retrieve relevant chunks
 7. Generate answer
 8. Decide whether to return answer or escalate
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3. Architecture Components

- Document Loader: Loads PDF
- Text Splitter: Creates chunks
- Embedding Model: Converts text to vectors
- Vector Database (ChromaDB): Stores embeddings
- Retriever: Finds relevant chunks
- Answer Generator: Generates response

- LangGraph Workflow: Controls flow
 - HITL Module: Handles escalation
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4. Workflow (LangGraph)

Nodes used:

- Retrieve
- Generate
- Decision
- HITL
- Output

Flow:

Query → Retrieve → Generate → Decision → Output / HITL

5. Conditional Logic

- If confidence ≥ 0.5 → return answer
 - If confidence < 0.5 → escalate to human
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6. HITL Mechanism

When the system cannot find relevant information:

- It asks for human input
 - Returns the human response as final output
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7. Design Choices

- Chunk size: 500 (balanced context)
 - ChromaDB: lightweight vector storage
 - HuggingFace embeddings: free and local
 - Rule-based answer generation: simple and stable
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8. Limitations

- No external LLM API used
 - Limited to one PDF
 - Basic answer generation
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9. Future Enhancements

- Add OpenAI/Groq API
 - Build UI (Streamlit)
 - Multi-document support
 - Add memory
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10. Conclusion

This project demonstrates a simple and effective RAG system with workflow control and HITL support, suitable for customer support automation.